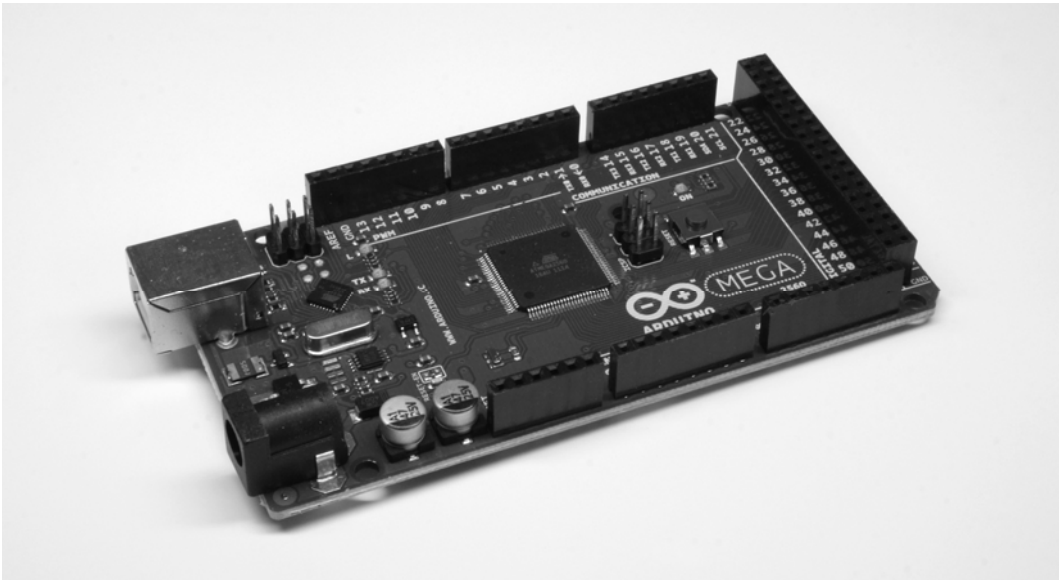


Figure 11-10. *Arduino Mega 2560*

The Mega 2560 is bigger than the little Uno in every way—it has 54 input and output pins, more than four times that of the Uno, 14 of which can be used for PWM; ten more analog inputs, for 16 total; four hardware serial ports instead of one; eight times the program memory at 256 kilobytes and four times the RAM and EEPROM space. Not only can it hold bigger sketches, it can, for example, control the speed of 14 DC motors, read from 16 different analog sensors, or even easily switch 128 LEDs with no additional hardware or tricky wiring needed. While there are always extra hardware components that can give the Uno similar levels of capabilities, the Mega 2560 doesn't need them, simplifying your wiring and your code.

Speaking of code, the Arduino Mega 2560 works pretty much just like its little brother. The pins used for I2C are in a different location and there are extra hardware serial ports, but otherwise the same code should work just fine on the Mega. The Mega is also compatible with most of the shields made for the Uno, including the Ethernet Shield, so the next time you need to blink a ton of LEDs, you might want to consider using the Arduino Mega 2560.

Learn Another Language

Now that you've got a decent handle on programming Arduino C, there is a world of new open source programming languages out there ready to be taken on. While altogether different from the Arduino's C-based language, we're going to have a quick glance over a couple of these programming languages that work really well with the Arduino platform and have complementary design ideologies. These languages will open up possibilities for using the Arduino to generate sound and images, manipulate video, and make things even more interactive. In addition to introducing the languages here, we will also have a look at a group of Arduino sketches called Firmata that make using inputs and outputs on the Arduino a whole lot easier.